Back to Data types

- Fundamental data type
  - int, char, float, bool, …
  - And arrays
- Custom data types
  - struct
    - Short for structure
  - class

struct : use case

- Need for grouping multiple different types
- Helps organize complex data in a meaningful way
- Examples ….
  - time : seconds, minutes, hours
  - date : day, month, year
  - book : author, title, price, year
  - player : username, experience
  - game leaderboard : player, rank, score

struct : definition

```
struct PlayerInfo
{
    char username[100];
    float experience;
};

struct GameLeaderboard
{
    PlayerInfo player;
    int rank;
    float score;
};
```

struct : variable declaration

```
struct PlayerInfo
{
    char username[100];
    float experience;
};

struct GameLeaderboard
{
    PlayerInfo player;
    int rank;
    float score;
};

GameLeaderboard resultItem
{
    {"skibidi",213.0},
    52,
    52.0
};
```

Structure : member access using . operator

```
struct Time
{
    //Use this format for readibility
    //one declaration on each line
    int seconds;
    int minutes;
    int hours;
};

Time createTime;

createTime.hours = 15;
createTime.minutes = 32;
createTime.hours = 11;

Time endTime{ 40,50,11};
```

Assingment

- Create a function to take a struct of PlayerInfo and print it.
- Create a function to
  - take 2 Time variables as parameters
  - Add them
  - And return the new Time
  - Hint
    - Time AddTime(Time timeA, Time time B){ //fill in here
    - }